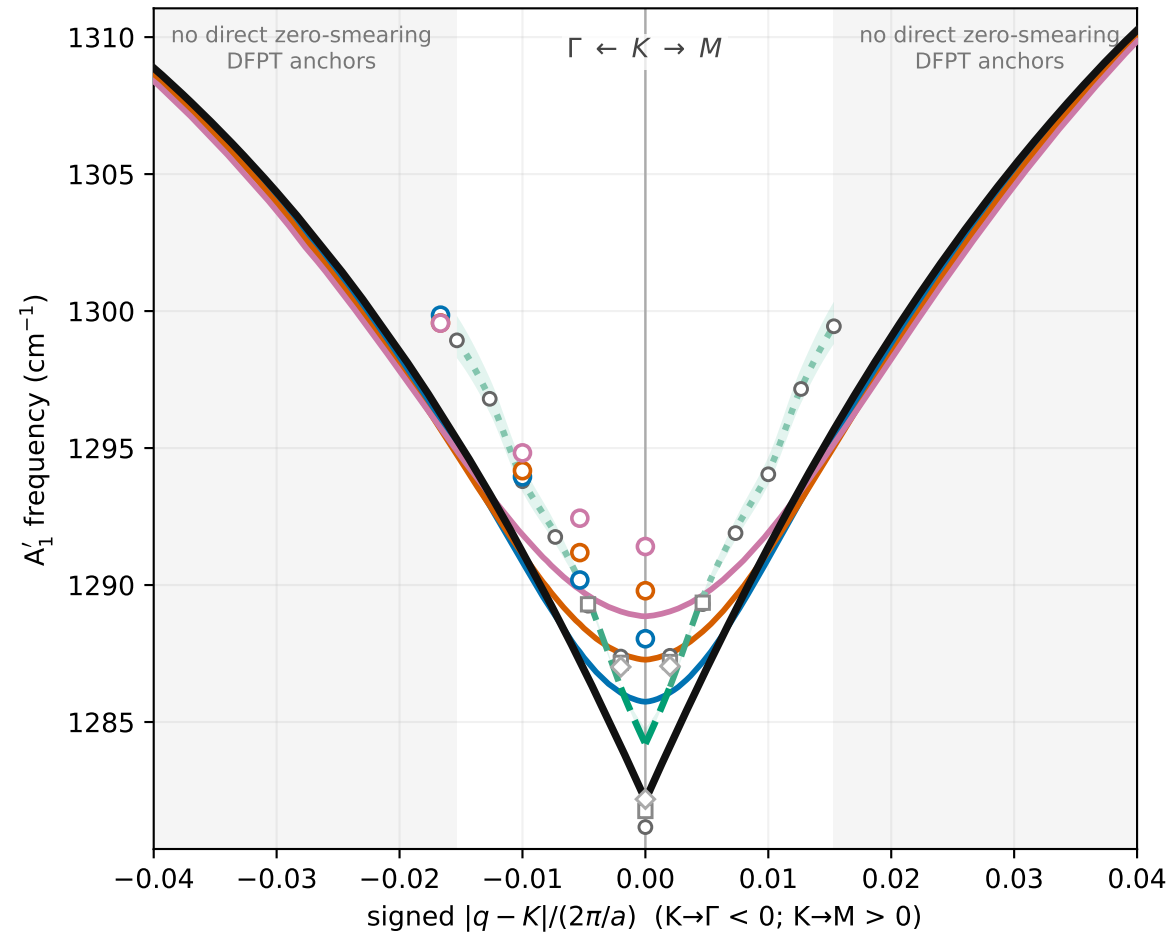
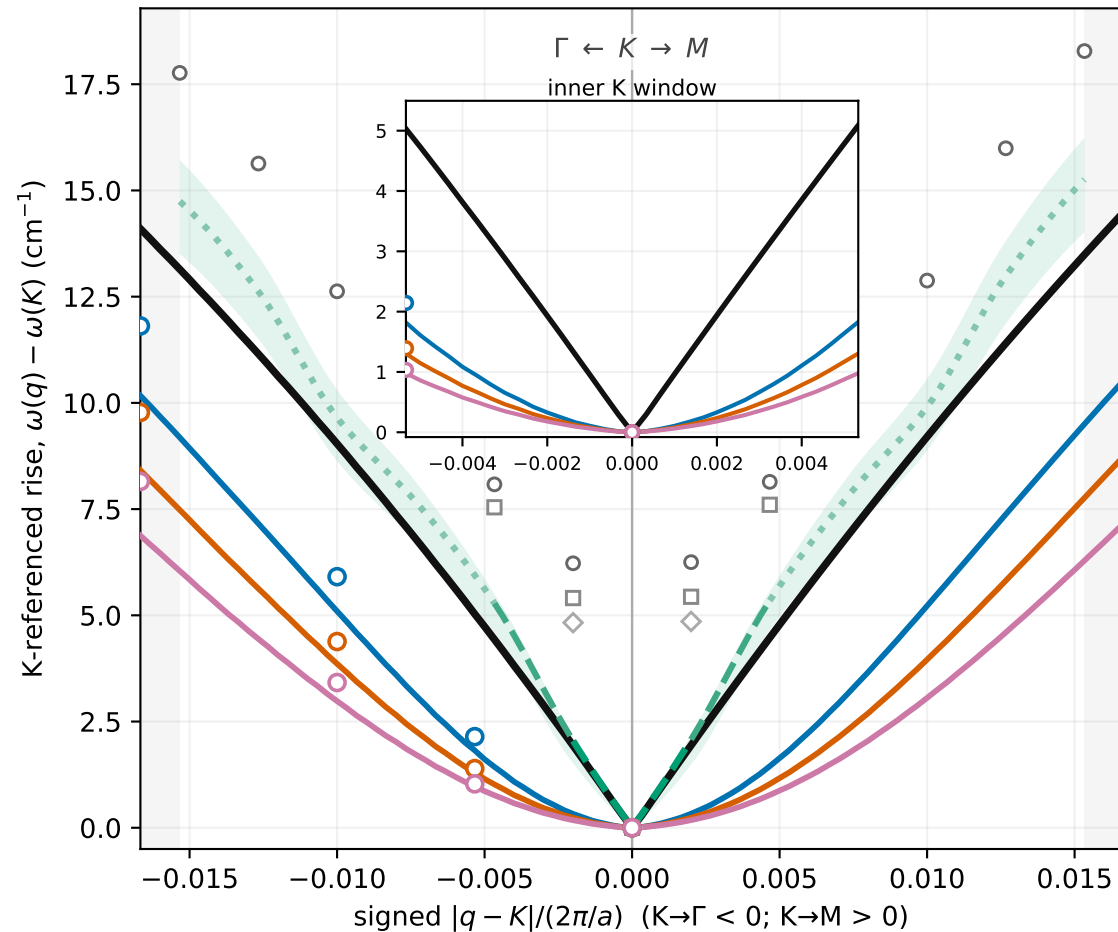


Graphene K-point A'_1 : electronic smearing at one fixed lattice condition

(a) Absolute frequency: extended K region



(b) Cusp shape: each series referenced to its own K



MLIP + full EPC LR
smearing/degauss (Ry)

- 0
- 0.0019001
- 0.0028501
- 0.0038002

- zero-smearing DFPT $N \rightarrow \infty$ guide
- zero-smearing guide envelope
- raw zero-smearing DFPT (k192/240/288)
- direct finite-smearing DFPT ($K \rightarrow \Gamma$ only)
- outside direct zero-smearing DFPT support

Lattice condition held fixed:
the same static short-range MLIP $D_{\text{SR}}(q)$
is used for all four curves. The paired
300/450/600 K lattice/TDEP backgrounds
from the source file are not used.

One common absolute EPC reference constant;
no per-smearing K re-anchoring. Finite-smearing
DFPT markers are direct static-lattice checks.